Adult Guide — E02 One Whole, Many Parts

Review version 1.0.0 Author: Maria Smith Audience: parents, carers, early-years practitioners and teachers Final-publication status: awaiting Maria Smith's approval

Purpose

The child rejoins Mia, Sol and Tavi immediately after E01. The parcel contains one Moon Lantern and a picture card asking the team to take it to the balcony. A small door blocks the way, and the lantern is too wide to fit through it. The book then holds one simple question in view: How can the whole lantern get through the small door?

Pax gives the child a four-step plan: check the whole, choose and count every part, carry every part through, then rebuild the same whole. Every activity is one necessary part of that plan. The vocabulary follows the experience rather than replacing it. The completed lantern reaches the balcony and naturally begins the two-label pattern that previews E03.

The permanent cast and Pax

Mia, Sol and Tavi remain the main trio. E02 introduces exactly one new guest: Pax, the patient piece-keeper. Pax checks fitted trays and asks process questions but does not identify the correct layout or count before an attempt.

Pax may return in a later book only when components, partitions or reassembly are relevant. A later callback may ask whether every place is filled or whether pieces overlap; it must not reveal a new answer.

Narrative cause chain

1. The E01 parcel supplies the Moon Lantern and picture card.
2. The small doorway creates the one practical problem.
3. The child identifies the original whole so they know what must be rebuilt.
4. The child chooses four complete, separate, same-size parts.
5. The child checks each carrying spot once so no part is missed or repeated.
6. All four parts pass through the doorway.
7. The child compares visible sizes and checks held and whole counts.
8. The child fills the gap and keeps the unrelated extra outside.
9. Every lantern part returns to the round frame; the tray is empty.
10. The child joins four separate one-part counts and chooses the exact total.
11. The four counted parts are fitted together as one whole lantern.

If the child loses the thread, point to the small door and say: “The whole lantern will not fit. What is our plan?” Then use the four short actions: “Check it. Take it apart. Carry every part. Build it again.”

Experience-before-vocabulary

- whole — all of this registered thing, with none of it missing and nothing

extra added;

- count — keep the complete generated row by visiting every item once,

skipping none and repeating none;

- complete partition — all of the agreed whole is covered by its registered

pieces;

- separate / non-overlapping — one piece does not sit over another piece;
- equal pieces — pieces made the same size in this particular partition;
- unequal pieces — visible pieces that are not the same size;
- held part — one or more chosen pieces contained in the complete

partition;

- held count — how many registered pieces are chosen; and
- whole count — how many registered pieces make the complete partition;
- plus / addition — join separate counted groups while keeping every source

item exactly once; and

- equals — both sides name the same exact total, here the same four visible

lantern parts.

Do not introduce parameter-free coordinate, fibre, candidate grammar, multiplicity or depth-independent closure to the child. Those exact terms remain here to protect the source boundary.

Exact SFT source and dependency order

1. Structural One

Claim: `SFT-FOUNDATION-ONE-001` Dependency: `SFT-ROOT-THERE-IS-NO-NOTHING` Status: independently replicated Provenance: forward forcing Closure: depth-independent Receipt: `sha256:68332624c276dc5d0ddfd568a26b95e20d7d5665fcdb7e825ff1b7b19bf1d51c`

> The unique minimal representation of the admitted root occurrence is its
> complete self-whole with no unforced addition; this structural identity is
> the One.

The complete generated product classifies root coverage as none, proper or complete, crossed with absence or presence of unforced extra material. Six forms are decided. Only `exact-self-whole` retains the complete admitted occurrence without omission or addition. Arithmetic and fractions are not imported into this claim.

2. Exact positive finite count

Claim: `SFT-FOUNDATION-COUNT-001` Dependency: `SFT-FOUNDATION-ONE-001` Status: independently replicated Provenance: forward forcing Closure: depth-independent across every generated nonempty finite extension Receipt: `sha256:4b2411716cbef2fb9fa7aac58b24ce5eb3c0c59fe7230fd703fd89b5260f6e23`

> Every generated nonempty finite succession of structural-One occurrences has one exact positive count representation: its complete generation trace, retaining every generated occurrence once and adding none.

The grammar has ten representation classes across coverage, retained multiplicity and ungenerated extra occurrences. Only `complete-coverage__once__no-extra` survives. E02 translates omission, duplication and addition into a four-light bridge. The bridge is a demonstration, not the formal proof.

3. Exact positive part coordinate

Claim: `SFT-FOUNDATION-PART-001` Dependency: `SFT-FOUNDATION-COUNT-001` Status: independently replicated Provenance: forward forcing Closure: depth-independent within the registered positive finite partition boundary Receipt: `sha256:87b40affae50458ba5492af91778fbca79a8de0bc3b951e520535093c2b7b728`

> For every registered finite partition of the structural One that is complete, disjoint and equal-fibred, and every nonempty held selection contained within it, the unique parameter-free exact part coordinate is the pair of positive counts identifying the held selection and the complete partition.

The formal grammar contains 192 generated alternatives. It classifies whole coverage, overlap, fibre equality, selection position, held-count presence, whole-count presence and extra coordinates. Exactly `complete__disjoint__equal__within__held-present__whole-present__no-extra` survives.

4. Exact arithmetic kernel

Claim: `SFT-MATH-EXACT-ARITHMETIC-001` Dependencies: exact positive count, exact part coordinate, part equivalence and form enforcement Status: independently replicated Provenance: forward forcing Closure: depth-independent Receipt: `sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418`

This downstream result admits exact arithmetic only after the generative count and part structures are sealed. Conventional symbols are correspondence labels after that derivation; they do not supply its authority.

5. Junction addition

Claim: `SFT-MATH-ARITH-JUNCTION-ADDITION-002` Dependencies: exact arithmetic, form enforcement and generated succession Status: independently replicated Provenance: forward forcing Closure: depth-independent Receipt: `sha256:ecbb32a37fb10022315429bb39d88f49437fc856396d0b8db42fca49b2032493`

> Addition is the complete junction of two held-disjoint generated traces, with every source unit retained exactly once.

E02 repeats that lawful junction to make the child-facing equation `1 + 1 + 1 + 1 = 4 parts`. The four source parts remain visible and are neither lost nor repeated. The later reassembly statement is deliberately written `4 parts → 1 whole lantern`, not `4 = 1`, because parts and whole lantern name different units.

The non-negotiable boundary

E02 does not derive or teach as SFT authority:

- numerical zero or an empty held selection;
- negative, imaginary, irrational, decimal or floating proof quantity;
- completed infinity or an ungenerated continuum;
- division imported as a foundational operation;
- fractions or the claim that each pictured part is a numerical quarter;
- equivalence between differently generated partitions; or
- the claim that every object called “one piece” has equal size.

The child-facing label `2 HELD · 4 MAKE THE WHOLE` keeps the exact positive pair. Do not silently replace it with a decimal. Do not claim it is interchangeable with `1 held · 2 make the whole`; cross-partition equivalence belongs to the separate admitted downstream theorem and is deliberately postponed.

How to share each challenge/reveal pair

1. Read the challenge and pause.
2. Let the child look across every pictured alternative.
3. Accept pointing, gaze, sign, speech, movement or communication aids.
4. Ask one process question without naming the answer.
5. Turn to the reveal and compare the same objects in the same positions.
6. Keep a first attempt that differed; identify exactly what changed.
7. Replay with safe large objects if useful.

The reveal is a shared check, not a score screen.

Page guidance and answers

Pages 3–8 — parcel, problem and plan

Pages 3–6 establish the previous-book callback, the one clear problem, the child-facing question, the four-step plan and the single new guest. Page 7 answer: the complete round lantern, not the lantern with a gap and not the complete lantern plus an attached extra. Page 8 introduces whole only after the child has compared all three.

Pages 9–13 — four carrying spots

Page 10 asks the child to visit the four visible spots once each. Page 12 preserves Sol's check `1, 2, 3, 3, 4`; the problem is the repeated third spot. Page 13 explicitly keeps that failed attempt and rebuilds the trace.

Pages 14–17 — choose the four-part plan

The accepted layout is the one that fills the entire agreed outline, has no overlap and uses same-sized pieces. A layout can fail more than one condition. Ask all three questions rather than accepting a familiar-looking picture. Pages 16–17 add the observable symmetry check only after the child has looked: left and right match around the middle, and top and bottom match around the middle. Symmetrical names that balanced match; it is not used as an unexplained instruction.

Pages 18–20 — carry every part

Every lantern part passes through the same small doorway. One carrying spot is used for each part so the child can see that none was missed or repeated.

Pages 21–22 — check the plan's sizes

Pair A is equal-sized; Pair B is visibly unequal. Unequal pieces are not called false or bad. The narrower claim is that counts alone cannot identify equal portions when the partition pieces are unequal.

Pages 23–25 — practice-frame check: gap and extra

The flat frame is a checking picture, not the final lantern rebuild. Page 23 is not complete because one registered place is unfilled. Page 24 returns the registered fourth part, then explicitly returns all four separate parts to the carrying tray. Page 25 keeps the extra triangle visible and outside; it does not disappear and is not absorbed into the lantern by a word. This keeps the story state coherent for the addition activity on pages 28–29 and the one final rebuild on page 30.

Pages 26–27 — held and whole counts

Answer: two pieces are held and four pieces make the complete registered partition. Both positive counts stay visible. Do not extend the activity to an empty held selection in this book.

Pages 28–29 — exact addition and equality

Answer: four parts. Invite the child to touch each separate part once before choosing the total. Read + as “plus” and explain that it joins the separate counts. Read = as “equals” and explain that both sides count the same four parts. Do not call the pieces quarters and do not hide or merge them while the equation is being checked.

Pages 30–32 — answer the story question

Page 30 changes from counting to reassembly: the exact four parts are fitted together as the same whole lantern. The arrow is a process arrow, not equality. Page 31 keeps the addition statement and the reassembly statement distinct. Page 32 previews the two-label transition and recurrence of E03 without teaching its result or unlocking Level Three.

Safe optional materials

The book is complete without materials. Optional use may include large card circles cut by an adult, four large foam shapes, a fitted tray outline and four large floor markers. Avoid small loose parts, sharp cutting tools in the child's reach, glass, hot lamps and flashing lights. The child may watch rather than handle pieces.

Accessibility

- Keep words and reveal labels above pictures.
- Read picture descriptions aloud when helpful.
- Use shape, position, outline and text; never colour alone.
- Offer tactile outlines and large pieces with distinct edge textures.
- Allow extra processing time and unlimited replay without penalty.
- Keep candidate positions stable between challenge and reveal.
- Use the semantic HTML edition for scalable text and structured headings.
- The PDF will contain selectable text but will not be called fully tagged

unless that is independently verified.

Companion Level Two requirements

Every paired stage must be directly playable by touch, keyboard and assistive technology and include a replay control. Mia, Sol and Tavi remain visible; Pax is the sole new Level Two guest. The final activity must require the child to place four visible parts, choose the total, read the equation and then rebuild the whole; a passive picture-memory sequence is not an acceptable substitute. The book-page bridge remains available after a correct answer. Optional codes unlock only character moments, alternate replay arrangements or an E03 preview.

Narration must be caption-matched and pre-rendered from Maria Smith's local Kokoro weights. Audio and sound effects are delivery tools only. The installed game must run offline, collect no personal or behavioural data and keep optional progress locally on the device.

Reference-only curriculum navigation

The activities can support early communication, counting actions, shape comparison, spatial language, matching, fine-motor movement, recall and collaborative checking. UK early-years frameworks calibrate presentation only; they do not supply or validate any SFT premise.

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 All story text, activities, layouts and later commissioned art are original to this series. Image and voice systems are production tools and have no scientific-authority role.